MILITARY SPECIFICATION

INTERCONNECTING GROUP,
ON-143(V)9/USQ

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for manufacture and acceptance of the Interconnecting Group, ON-143(V)9/USQ, hereinafter referred to as the IG.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-5015	Connectors, Electrical, AN Type, Circular Threaded, General Specification For
MIL-E-16400	Electronic Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts), Packaging Of
MIL-C-26482	Connectors, Electrical, (Circular, Miniature, Quick Disconnect Environment Resisting), Receptacles And Plugs, General Specification For
MIL-I-46058	Insulating Compound, Electrical (For Coating Printed Circuit Assemblies)
MIL-C-55543	Cable, Electrical Flat Multiconductor, Flexible, Unshielded

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-12	Abbreviations For Use On Drawings, And In Specifications, Standards And Technical Documents
MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-118-114	Electrical Characteristics Of Digital Interface Circuits
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-003-12), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5820

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MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-740	Airborne And Structureborne Noise Measurements And Acceptance Criteria Of Shipboard Equipment
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-965	Parts Control Program
MIL-STD-1397	Input/Output Interfaces, Standard Digital Data, Navy Systems
DoD-STD-1399, Section 070, Part 1	Interface Standard For Shipboard Systems, D.C. Magnetic Field Environment (Metric)
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electrical Power, Alternating Current (Metric)
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

HANDBOOKS

MILITARY

MIL-HDBK-237	Electromagnetic Compatibility Management Guide For Platforms, Systems And Equipment
MIL-HDBK-241	Design Guide for Electromagnetic Interference (EMI) Reduction In Power Supplies
MIL-HDBK-251	Reliability/Design Thermal Applications

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)

55910-0104137	Circuit Card Assembly, Dynamically Adaptive Receiver Transmitter
55910-0104140	Circuit Card Assembly, Radio Interface Buoy
55910-0104143	Circuit Card Assembly, NTDS
55910-0104146	Panel Assembly, Front
55910-0104150	Chassis Assembly
55910-0104151	Cage Assembly, Circuit Module
55910-0104152	Cage Assembly, BLACK Circuit Module
55910-0104153	Control Indicator, C-10582A/USQ Data List, Control Indicator,
55910-0104157	Power Supply, 5 VDC Or ±12 VDC
55910-0104160	Interconnecting Group ON-143(V)6/USQ
55910-0104604	Circuit Card Assembly, RED/BLACK Interface
55910-0104642	Panel, Power Distribution
55910-0104645	Panel Assembly, Rear
55910-0105714	Cable Assembly, BLACK I/O (1W1)
55910-0105715	Cable Assembly, BLACK I/O (1W2)
55910-0105716	Cable Assembly, BLACK I/O (1W3)
55910-0105717	Cable Assembly, NTDS Input (1W4)
55910-0105718	Cable Assembly, NTDS Ouput (1W5)
55910-0105719	Cable Assembly, HSP (1W6)
55910-0105720	Cable Assembly, PTO (1W7)
55910-0105721	Cable Assembly, VOC, REP (1W8)
55910-0105722	Cable Assembly, KG, TTY (1W9)
55910-0105723	Cable Assembly, Jumper (1W10)
55910-0105724	Cable Assembly, Power Supply (1W11)
55910-0105725	Cable Assembly, SEN (1W12)
55910-0105726	Cable Assembly, Blower (1W13)
55910-0105727	Cable Assembly, RED/BLACK, PWR-SIG (1W14)
89199-0008122	Circuit Card Assembly, Processor Control 1
89199-0011097	Circuit Card Assembly, Memory, Read Only, Reprogrammable 3

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PUBLICATIONS

NATIONAL SECURITY AGENCY (NSA)

NACSIM 5100A	Compromising Emanations Laboratory Test Stand, Electromagnetic
NACSIM 5110	Compromising Emanations Laboratory Test Stand, Electromagnetic

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

WS-6536D	Procedures And Requirements For Preparation And Soldering Of Electrical Connections

SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)

EE130-AC-OMI-010/E 1060N143V6	Operation And Maintenance Instructions For Interconnecting Group (IG) ON-143(V)6

ASSISTANT SECRETARY OF THE NAVY (SUPPLY AND LOGISTICS)

NAVMAT P4855-1	Navy Power Supply Reliability, Design And Manufacturing Guidelines
TADSTAND C	NAVMAT Letter MAT 08Y/OLM Ser 232, 2 July 1980

NAVAL MILITARY PERSONNEL COMMAND (NMPC)

NAVPERS 18068	Manual Of Navy Enlisted Manpower And Personnel Classifications And Occupational Standards

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extend specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA RS-310-C-77	Racks, Panels And Associated Equipment

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The IG shall be in accordance with MIL-E-16400 to the extent specified herein.

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3.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Parts, materials, and processes. The design practices, parts, processes, and materials developed and procured for use in the IG shall ensure fulfillment of all requirements of this specification.

3.2.1 Selection. Selection of parts, materials, and processes shall conform to the Selection paragraph of MIL-E-16400.

3.2.1.1 Used or damaged parts or materials. Used or damaged parts or materials shall not be used.

3.2.1.2 Interchangeability. Interchangeability shall conform to the Interchangeability paragraph of MIL-E-16400. Drawing (DWG) 55910-0104160 shall be the standard for measuring interchangeability of all applicable units, assemblies, subassemblies, and parts.

3.2.2 Parts. Parts shall be as specified in 3.2.2.1 through 3.2.2.9.

3.2.2.1 Parts control. Parts control shall be in accordance with MIL-STD-965, Procedure I or Procedure II, as applicable.

3.2.2.2 Standard parts. Standard parts shall conform to the Standard parts paragraph of MIL-E-16400.

3.2.2.3 Nonstandard parts. Nonstandard parts shall conform to the Nonstandard parts paragraph of MIL-E-16400.

3.2.2.4 Part application. The operating temperature of heat dissipating and heat sensitive parts shall be within the thermal stress specified in the reliability prediction report. Heat sensitive parts shall be thermally isolated from heat sources by one or more of the methods specified in a through d:

a. Physical separation
b. Use of thermal insulation techniques
c. Placement of such parts upstream in the cooling fluid or path, and heat sources downstream
d. Maximum use shall be made of conduction techniques to remove heat from heat dissipating parts along short metallic paths of large cross-sectional area to the heat sink or exchange.

3.2.2.5 Air filters. The forced air cooling inlet and outlet temperature differential shall not exceed 14° Celsius (C). Filters shall be provided and located at all air intakes. Filters shall be readily removable for cleaning without disassembly of the equipment.

3.2.2.6 Capacitors. Capacitors shall be selected in accordance with the Capacitors paragraph of MIL-E-16400.

3.2.2.6.1 Capacitors, fixed, dry electrolytic (aluminum foil). The use of fixed, dry electrolytic capacitor shall be in accordance with the Capacitors, fixed, dry electrolytic (aluminum foil) paragraph of MIL-E-16400.

3.2.2.6.2 Capacitors, fixed, paper or paper-plastic (nonmetallic case). Fixed, paper or paper-plastic capacitors with nonmetallic cases shall be in accordance with the Capacitors, fixed, paper or paper-plastic (nonmetallic case) paragraph of MIL-E-16400.

3.2.2.6.3 Mica-dielectric capacitors. The use of mica-dielectric capacitors shall be in accordance with the Mica-dielectric capacitors paragraph of MIL-E-16400.

3.2.2.6.4 Capacitors, fixed, paper dielectric. Fixed paper dielectric capacitors shall not be used.

3.2.2.7 Microelectronic devices. The selection and application of microelectronic devices shall conform to the Microelectronics paragraph of MIL-E-16400.

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3.2.2.8 Connectors. All connectors shall conform to MIL-C-26482 or MIL-C-5015, as shown in DWG 55910-0104160.

3.2.2.8.1 Mating connectors. A complete set of mating halves of connectors, backshells and protective caps shall be provided. Except for coaxial types, mating connectors shall be solderless (crimp-on) type whenever possible. Connectors shall be selected, designed and installed so they cannot be mismatched or cross-connected.

3.2.2.9 Switches. All electrically actuated mechanical switches, except momentary switches, built-in-test (BIT) and built-in-test equipment (BITE) switches, and relays shall be the self-latching type, and shall not require power to remain in any selected position.

3.2.3 Materials. All material used in the construction and fabrication of the IG shall be in accordance with the Materials paragraph of MIL-E-16400. The requirement for nonmagnetic material is excluded. The contractor shall certify that the equipment materials used will not be of toxic or hazardous substances and that the material shall have no adverse effect on the health of personnel when used for its intended purposes.

3.2.3.1 Arc-resistant materials. Arc-resistant materials shall conform to the Arc-resistant materials paragraph of MIL-E-16400.

3.2.3.2 Insulating materials, electrical. Electrical insulating materials shall conform the Insulating materials, electrical paragraph of MIL-E-16400.

3.2.3.3 Dissimilar metals. The selection and application of materials used in design and fabrication shall conform to the requirements of the Dissimilar metals paragraph of MIL-E-16400.

3.2.3.4 Ferrous alloys, corrosion resistant. Ferrous alloys shall conform to the Ferrous alloys, corrosion resistant, paragraph of MIL-E-16400.

3.2.3.5 Aluminum alloys. Aluminum alloys shall be in accordance with the Lightweight metal (aluminum alloy) paragraph of MIL-E-16400.

3.2.3.6 Other metals. Nonferrous metals, except aluminum, shall be in accordance with the Non-ferrous metals paragraph of MIL-E-16400.

3.2.3.7 Unacceptable materials. Equipment manufactured in accordance with this specification shall not include polychlorinated biphenyls, asbestos and asbestos compounds, fragile or brittle materials, beryllium and beryllium compounds unless so identified, and lithium and lithium compounds not specifically approved by the procuring agency. The Unacceptable materials paragraph of MIL-E-16400 shall apply.

3.2.2.8 Adhesives. Adhesives shall conform to the Adhesives paragraph of MIL-E-16400.

3.2.2.9 Lubricants. Lubricants shall conform to the Lubricants paragraph of MIL-E-16400.

3.2.3.10 Fungus. The equipment shall be resistant to fungi in the operating and nonoperating condition and shall not be adversely affected when tested in accordance with the Fungus paragraph of MIL-E-16400.

3.2.4 Processes. Processes shall be as specified in 3.2.4.1 through 3.2.4.4.2.

3.2.4.1 Soldering. Soldering shall conform to WS-6536D. Soft soldering shall not be used for mechanical support.

3.2.4.2 Structural welding. Structural welding shall conform to the Structural welding paragraph of MIL-E-16400.

3.2.4.3 Castings. Castings shall be in accordance with the Castings paragraph of MIL-E-16400.

3.2.4.4 Painting. Only the exterior surface of the cabinet shall be painted.

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3.2.4.4.1 Surface preparation. Surface preparation shall conform to the Painting paragraph of MIL-E-16400.

3.2.4.4.2 Color. Paint shall be gray enamel, Color No. 26307, in accordance with FED-STD-595. Preparation and painting shall conform to the Painting paragraph of MIL-E-16400.

3.3 Design. The IG shall be manufactured in accordance with DWG 55910-0104160 and this specification. The manufacturer shall provide equipment that is pin-for-pin compatible and interchangeable to the lowest replaceable unit (LRU) of existing IGs. Proposed changes, either to correct DWG discrepancies or to effect improvements, shall be approved by the procuring activity prior to their incorporation.

3.3.1 IG description. The IG shall maintain TEMPEST isolation between RED/BLACK security classifications and perform the functions specified in a through h:

a. Generate timing and delays
b. Provide a TD-1271( ) DAMA Modem Data Interface
c. Provide a TD-1271( ) Remote Request Interface
d. Provide vocoder synchronization
e. Perform all the necessary Link functions to maintain satellite communications between either NAVMACS/CUDIXS via a TD-1271 DAMA Modem
f. Perform all the necessary Link functions to maintain satellite communications between either NAVMACS/CUDIXS in a nonDAMA net
g. Initiate the self and system diagnostic tests specified in 1 through 6:
    1. Power-on BITE
    2. Crypto loop test
    3. SAT loop test
    4. Voice loop test
    5. Control panel test
    6. TD-1271 loop test
h. By means of front mounted control panel, allow the operator to select mode of operation, enter operating parameters, verify and enter data by touching the appropriate indicator. For ease of operation, the front-mounted control panel shall be capable of being remotely located. The processor shall provide the control required to interact with the operator and to allow the use of the same suite of equipment for these communications networks. The IG shall be a standard 19-inch (in.) rack-mountable unit with an enclosed chassis. All controls and indicators necessary for operation shall be located on the front panel, although remote location of the control panel shall be possible. A cover shall be provided for the IG when the control panel is in a remote location. All input, output, and power connectors shall be mounted on the rear panel. The IG front panel and chassis shall be mounted on slide rails to allow the IG to be extended from its case for access to the card cages and internal circuitry. The IG shall be broken down into the basic areas specified in 1 through 5:
    1. Front panel, rear panel, and case
    2. RED card cage
    3. BLACK card cage
    4. RED/BLACK interface
    5. Power supplies.

FIGURE 1 is an interface block diagram of the IG and system equipment. FIGURE 2 is an overall functional block diagram of the IG.

3.3.1.1 Front panel, rear panel, and case. The front panel shall contain the operating controls and indicators and the control panel. An internally mounted fan and radio frequency interference (RFI) attenuation air filter shall be used to cool the IG. When the front panel is mated with the housing, an RFI gasket on the front panel shall ensure RFI integrity. The front end chassis shall be capable of extension by loosening knurled fasteners at each corner and pulling outward on the handles on both sides of the front panel until the detents in the side rails engage. With the unit so extended, the RED and BLACK card cages, power supplies, and internal circuits shall be accessible. Flat cables with plug-in connectors shall provide the interface between the rear panel connectors and the IG circuitry.

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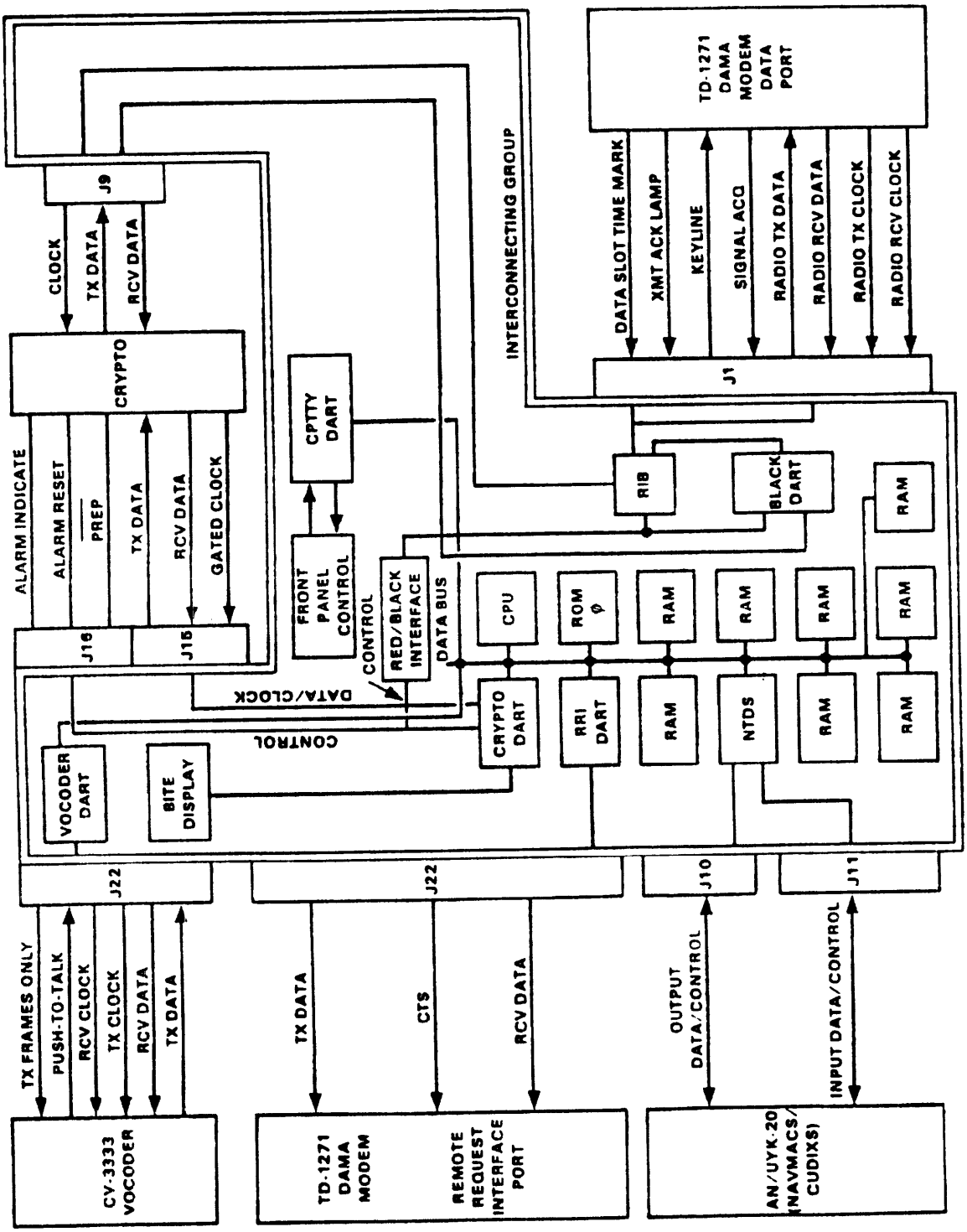


FIGURE 1. Detailed interface block diagram.

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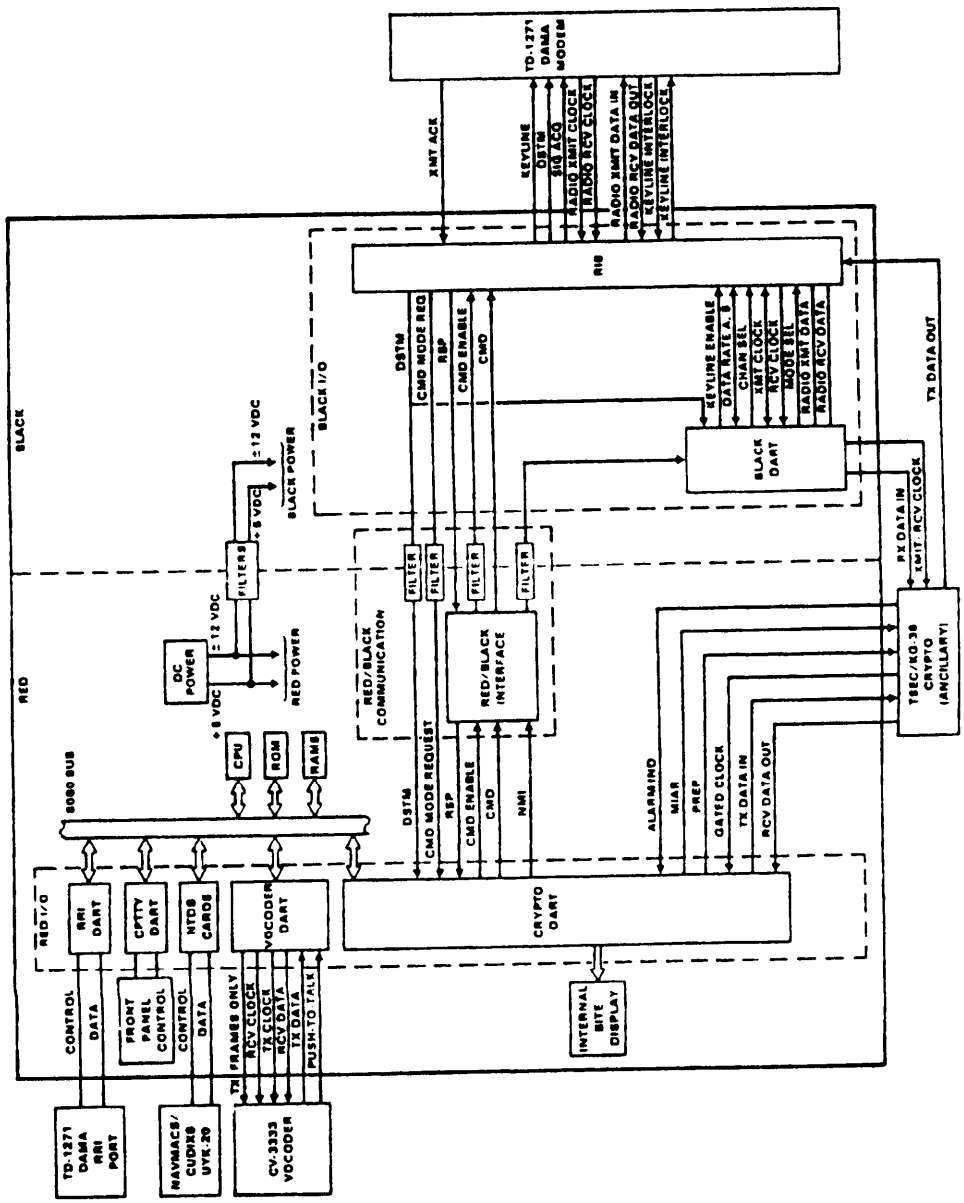


FIGURE 2. Overall functional block diagram.

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3.3.1.2 RED card cage. The RED card cage shall be mounted to the chassis and shall be extended out of the case with the front panel. The RED card cage shall contain the 16 circuit cards that perform the various classified system functions. The RED card cage shall contain the cards specified in a through e:

a. One central processing unit (CPU)
b. Eight random access memory (RAMs)
c. One read only memory (ROM)
d. Four dynamically adaptive receiver transmitter (DARTs):
   1. One Vocoder DART
   2. One Crypto DART
   3. One CPTTY DART
   4. One remote request interface (RRI) DART
e. Two Naval Tactical Data System (NTDS) interface cards

Circuitry from the RED card cage to the rear panel shall be in the form of flat multiconductor ribbon cables. The RED card cage end of the ribbon cables shall terminate with 40-pin male strip plugs. The rear panel end of the ribbon cables shall terminate with round multipin connectors on the rear panel. The flexibility of the ribbon cables shall allow the chassis to be extended on the slide rails.

3.3.1.3 BLACK card cage. The BLACK card cage shall be mounted to the front panel and slide in and out along with the RED card cage. The BLACK card cage shall contain two circuit cards that perform the various unclassified system functions. To provide TEMPEST isolation, the BLACK card cage shall be physically separated and electronically filtered from the RED card cage. Connections from the BLACK card cage to the rear panel shall be accomplished via ribbon cables as specified in 3.3.1.2. The BLACK card cage shall contain the cards specified in a and b:

a. One BLACK DART
b. One radio interface board (RIB)

3.3.1.4 RED/BLACK interface. Signals passing through the RED/BLACK interface card shall either be filtered by low-pass electrical filters or isolated by solid-state relays. The solid-state relays shall be located on the RED/BLACK interface card, housed on the RED chassis next to the RED/BLACK transition. The low-pass filters shall be chassis-mounted. All interconnecting wiring through the RED/BLACK partition shall be shielded.

3.3.1.5 Processor bus structure. The address selection shall be controlled by the CPU and shall be routed to all other processor cards on a common 16-line address bus. Control lines between the processor cards shall be routed on common bus lines. Data shall be transferred among the cards on an eight-line bidirectional data bus.

3.3.1.6 Programming language. A Navy Standard High Order Programming Language (HOL) as specified in Tactical Digital Standard (TADSTAND C) shall be utilized. Deviations from the use of those Navy standard HOLs shall require prior written approval to a waiver request in accordance with TADSTAND C.

3.3.1.7 Power supplies. Two power supplies shall be located next to the RED card cage and shall supply the regulated +5 volts-direct current (VDC) and ±12 VDC used for IG operation. Power supplies shall conform to or exceed the requirements specified in NAVMAT P-4855.1.

3.3.2 Interface definition. The IG shall interconnect the crypto, TD-1271( ) (data and RRI Ports), audio encoder, and AN/UYK-20 processor in fleet satellite communication (FLTSATCOM) system installation (FIGURE 1).

3.3.2.1 Data interface. The IG shall be capable of interfacing on the communications channels and at the data rates specified in TABLE I. All input/output (I/O) signals specified in TABLE I shall be in conformance with the Unbalanced voltage digital interface circuit paragraph of MIL-STD-188-114 for ±6 volt (V) unbalanced lines with the exception of the RRI and NTDS interfaces which shall be in conformance with MIL-STD-1397.

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TABLE 1. ON-143(V)9/USQ I/O channels.

(I/O) boards	Port ID 1/	Description	Transfer data rate (bps)
Vocoder DART	J22	Digital-coded voice	2400
CPTTY DART	J18	Control panel	2400
RIB	J1	Data for radio transmission	2400
BLK DART	J9	Encrypted data to crypto	2400
RIB	J9	Encrypted data from crypto	2400
Crypto DART	J15/16	KG-36 interface	2400
RRI DART	J22	RRI	9600
NTDS	J10/11	Interprocessor control	83.3 kbps 2/
CPTTY	J21 panel	Remote control	2400

1/ Identification
2/ Kilobits per second

3.3.2.2 Electrical power interface. The IG shall operate from a single-phase, 60 hertz (Hz), 115 volt alternating current (VAC) power source which provides Type I power as specified in DoD-STD-1399, Section 300 and shall comply with the requirements and constraints of that document and NAVMAT P-4855.1. The IG shall require not more than 250 watts power.

3.3.2.2.1 Application of BLACK power. The IG shall be capable of direct application of unfiltered (BLACK) power without compromising emanations being present on the power line.

3.3.2.4 Strappable options. Provision shall be made for the selection of the options specified in a and b (see 6.2) by backplane strapping:

a. NTDS fast I/O selection
b. Selection of teletype (TTY) data rates: 75 bits per second (bps), 300 bps, 1200 bps or 2400 bps

3.3.3 Major component list. The IG shall consist of the components specified in TABLE II. Component parts common to the IG shall be fully interoperable and interchangeable with previously manufactured IG components to facilitate interoperability, interchangeability, training, and logistics.

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TABLE II. Major component list.

Component	DWG No.
Interconnecting Group (1)	55910-0104160
Panel Assembly, Front (1A1)	55910-0104146
Control Indicator C-10582A/USQ (1A1A1)	55910-0104153
Panel Assembly, Rear (1A2)	55910-0104645
Cage Assembly, BLACK Circuit Module (1A3)	55910-0104152
Cage Assembly, RED Circuit Module (1A4)	55910-0104151
Panel, Power Distribution (1A5)	55910-0104642
Circuit Card, RED/BLACK Interface (1A6)	55910-0104604
Circuit Card, BLACK DART (1A3A3)	55910-0104137
Circuit Card, RIB (1A3A6)	55910-0104140
Circuit Card, Vocoder DART (1A4A32)	55910-0104137
Circuit Card, RRI DART (TBD)	55910-0104137
Circuit Card, CPTTY DART (1A4A30)	55910-0104137
Circuit Card, Data Link DART (1A4A29)	55910-0104137
Circuit Card, Crypto DART (1A4A28)	55910-0104137
Circuit Card, RAM (1A4A20) (C)	89199-0011101
Circuit Card, RAM (1A4A18) (A)	89199-0011101
Circuit Card, RAM (1A4A17) (9)	89199-0011101
Circuit Card, RAM (1A4A16) (8)	89199-0011101
Circuit Card, RAM (1A4A15) (B)	89199-0011101
Circuit Card, RAM (1A4A19) (3)	89199-0011101
Circuit Card, RAM (1A4A14) (2)	89199-0011101
Circuit Card, RAM (AA4A13) (7)	89199-0011101
Circuit Card, NTDS (1A4A7) (0-7)	55910-0104143
Circuit Card, NTDS (1A4A8) (8-15)	55910-0104143
Circuit Card, ROM (1A4A12) (0)	89199-0011097
Circuit Card, CPU (1A4A11)	89199-0008122
Power Supply (1PS1)	55910-0104157-1
Power Supply (1PS2)	55910-0104157-2
Cable Assembly (1W1)	55910-0105714
Cable Assembly (1W2)	55910-0105715
Cable Assembly (1W3)	55910-0105716
Cable Assembly (1W4)	55910-0105717
Cable Assembly (1W5)	55910-0105718
Cable Assembly (1W6)	55910-0105719
Cable Assembly (1W7)	55910-0105720
Cable Assembly (1W8)	55910-0105721
Cable Assembly (1W9)	55910-0105722
Cable Assembly (1W10)	55910-0105723
Cable Assembly (1W11)	55910-0105724
Cable Assembly (1W12)	55910-0105725
Cable Assembly (1W13)	55910-0105726
Cable Assembly (1W14)	55910-0105727

3.3.4 Mechanical design. Mechanical design shall be as specified in 3.3.4.1 through 3.3.4.5.5.

3.3.4.1 Enclosures. The equipment enclosures shall conform to the Enclosure paragraph of MIL-E-16400 and 3.3.4.2.

3.3.4.2 Degree of enclosure. The IG enclosure shall be dripproof (45 degrees) in accordance with MIL-STD-108.

3.3.4.3 Equipment slide and tilt mechanisms. The equipment shall have slide mechanisms of suitably rugged construction to support the drawer in the withdrawn position. The slide mechanism shall slide with a minimum of sliding friction without a tendency to bind. The slide mechanism shall be constructed of corrosion-resistant material, such as corrosion-resistant steel. Guide support shall be provided between the rear of each chassis and the enclosure to prevent damage, due to shock, to the chassis slides.

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3.3.4.4 Shock mounting. All slide-mounted equipment shall include shock pins at the rear of the unit.

3.3.4.5 Controls, indicators, and panel layouts. The design of controls and indicators, and their arrangement on operator and maintainer panels, shall conform to the criteria of MIL-STD-1472 and shall be in accordance with the Controls, indicators, and panel layout paragraph of MIL-E-16400. The color coding and marking of pin jacks and other test points shall be in accordance with MIL-STD-415. Abbreviations used on panels shall be in accordance with MIL-STD-12.

3.3.4.5.1 Front panel layout. Front panel layout of cabinets and drawers shall include consideration of sequencing procedures to facilitate performance by Navy electronic technicians, second class (ET-2) for setup operation and maintenance. All controls and metering necessary for normal operation of the IG shall be provided on the IG front panel.

3.3.4.5.2 Alarm lamps. Lamps used on front panels for alarms and other purposes shall conform to the requirements of the Transilluminated Displays paragraph of MIL-STD-1472. A capability for remote indication of the status of the alarm lamps shall be provided. Legend lights shall be used in preference to simple indicator lights, except where design considerations demand that simple indicators be used. Lamps used in legend switches shall conform to the requirements of the Legend Switches paragraph of MIL-STD-1472. Legend switches shall be used in preference to toggle switches, except where design considerations demand that toggle switches be used. Lamp testing shall be in accordance with the Lamp Testing paragraph of MIL-STD-1472.

3.3.4.5.3 Protection from improper operation. The IG design shall include features to protect the equipment from improper operational procedures and practices.

3.3.4.5.4 Chain reaction failures. To the extent practicable, the IG protective design shall inhibit chain reaction failures in which a failure in one component results in damage to other components.

3.3.4.5.5 Locking devices. Locking devices shall be provided on all critical internal adjustment controls to prevent the need for readjustment of controls after sustained periods of shock and vibration.

3.3.5 Airborne and structureborne noise. The IG shall be in conformance with the requirement limits of MIL-STD-740 for Grade A, Type 3 equipment and as specified herein. The maximum allowable sound pressure level for airborne noise shall be 60 decibel amperes measured at meter distance from the equipment. The structureborne noise criteria shall be 65 decibels (dB) for one-third wave band frequencies up to 800 Hz. Both the airborne and structureborne noise criteria shall be applicable when the equipment is operating normally.

3.3.6 Thermal design. IG thermal design shall be as specified in 3.3.6.1 and 3.3.6.2 and in accordance with the Thermal design paragraph of MIL-E-16400. Air cooling shall be used. The design of the equipment shall allow installation in open racks.

3.3.6.1 Heat removal. The heat removal method shall remove at least 80 percent of the heat generated within the equipment. MIL-HDBK-251 may be used as guidance in the design and selection of the heat removal method.

3.3.6.2 Cooling during maintenance. Cooling during maintenance shall conform to the Cooling during maintenance paragraph of MIL-E-16400.

3.3.7 Prime power design. Prime power design shall be as specified in 3.3.7.1 through 3.3.7.6.2.

3.3.7.1 Performance. The IG shall conform to the requirements of 3.8 when operated as specified in 3.3.6.2 through 3.3.6.6.2, in accordance with DoD-STD-1399, Section 300, Type I. Power supply reliability shall comply with the requirements of NAVMAT P4855-1.

3.3.7.2 Prime power voltage (low-level equipment). Prime power voltage for low level equipment shall be 115 VAC, 60 Hz, single-phase.

3.3.7.3 Prime power control. A prime power circuit breaker shall be provided in the IG cabinet. The circuit breaker shall conform to the Circuit breakers paragraph of MIL-E-16400.

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3.3.7.4 Corona and electrical breakdown prevention. Corona and electrical breakdown prevention shall conform to the Corona and electrical breakdown prevention paragraph of MIL-E-16400.

3.3.7.4.1 Insulation resistance. IG electrical circuits shall be capable of withstanding the insulation resistance tests of the Insulation resistance paragraph of MIL-E-16400.

3.3.7.5 Bonding and grounding. Bonding and grounding shall conform to the Bonding and grounding paragraph of MIL-E-16400.

3.3.7.6 Internal wiring and cabling. Internal wiring and cabling shall be in accordance with the Internal wiring and cabling paragraph of MIL-E-16400.

3.3.7.6.1 Flat multiconductor cable. Flat multiconductor cable shall be selected from MIL-C-55543.

3.3.7.6.2 Internal wiring hookup. The selection and application of internal electrical hookup shall conform to MIL-STD-454, Requirement 20.

3.3.8 Printed wiring. Printed wiring shall conform to the Printed wiring paragraph of MIL-E-16400.

3.3.8.1 Printed wiring board dimensions. Printed wiring board dimensions shall be as specified in DWG 55910-0104160.

3.3.8.2 Conformal coating. Printed circuit (PCs) cards shall be conformal coated to conform to MIL-I-46058.

3.3.9 Maintenance design. Maintenance design shall conform to the Maintenance design paragraph of MIL-E-16400.

3.3.9.1 Self-test provisions. The IG shall use BIT and BITE, or both, to:

a. Verify that the on-line equipment is operating properly
b. Fault located to the subassembly level when the equipment is on-line or off-line

Maintenance assistance modules may supplement BIT and BITE.

3.3.9.2 Accessibility. The IG shall be designed to provide accessibility to all limited life components. The IG shall conform to the Accessibility paragraph of MIL-E-16400.

3.3.9.3 Maintenance concept. The maintenance concept for correction of equipment malfunctions at the organizational level shall be the removal and replacement of LRUs to restore the equipment to conformance with operating requirements specified herein. LRUs removed at this maintenance level will be limited to plug-in modules, PC cards, or chassis-mounted parts which are readily isolated by BITE and standard shipboard general purpose test equipment. Complexity of LRUs should be kept to a level such that LRU repair can be accomplished at the intermediate maintenance level by Navy electronics technician, second class (ET-2) personnel with skills no greater than those specified in NAVPERS 18068 for a ET-2.

3.3.9.4 Modular construction. The principles and techniques applicable to modular design and construction shall be as specified herein and in accordance with the Modular construction paragraph of MIL-E-16400.

3.3.9.5 Circuit test point design criteria and test procedures. Circuit test point accesses which comply with the criteria specified in MIL-STD-415 shall be provided.

3.3.9.6 Special tools. The IG shall be serviceable with a minimum of special tools (see 6.5). All special tools needed shall be supplied. Special tools shall be subject to approval by the Government and shall be in accordance with the Special tools paragraph of MIL-E-16400.

3.3.9.7 Power receptacles. Convenience power receptacles shall not be provided.

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3.3.9.8 Hinged covers, panels, and access doors. Hinged covers, panels, and access doors, shall conform to the Hinged covers, panels, and access doors paragraph of MIL-E-16400.

3.3.9.9 Part protection. The IG shall be so designed that parts will not be susceptible to damage during equipment servicing and maintenance.

3.3.9.10 Protective caps. Protective caps shall conform to the Protective caps paragraph of MIL-E-16400.

3.4 Transportability. After preparation for delivery as specified in Section 5, the IG or modules, or both, shall be transportable by air, rail, ship, or surface carrier, without damage.

3.5 Performance characteristics. Performance characteristics shall be as specified in 3.5.1 through 3.5.2.2.

3.5.1 Functional description. The IG is an integral part of any CUDIX/NAVMACS configuration (ashore or afloat). The IG performance, in its selected configuration, is controlled by processor programs and IG electronic circuitry. The functional description specified in 3.5.1.1 through 3.5.1.6 cover the major functions and signal flow within the IG.

3.5.1.1. General. The IG shall provide the means for performance of major system functions specified in a through g and in 3.5.1.2 through 3.5.1.7:

a. Link control
b. Operator interface
c. Message data handling
d. Signal interfacing
e. Automated event sequencing
f. Secure voice operation
g. Loopback and BITE testing

3.5.1.2 Link control. Link control shall exercise the links protocol pattern established to conform to the operational needs of subscribers. All RRI protocol shall be generated and received via the RRI interface DART. A communications interface shall be established and maintained between the IG and the DATA processor. Link data transmitted and received shall be formatted and passed via the interprocessor channel.

3.5.1.3 Operator interface. Operator interface shall be exercised via the IG operator control panel. By selecting distinct modes on the control panel, the operator shall be able to enter commands and parameters for controlling system operations. The control panel shall display systems status information and alerts, and provide error and fault indications for system level and IG level troubleshooting.

3.5.1.4 Signal interfacing. Components interfacing with the IG generate and require signals with differing characteristics. The IG shall convert these signals to the required form for system compatibility.

3.5.1.5 Automated event sequencing. Automated event sequencing provide for automatic system operation by program controlled gating and timing logic.

3.5.1.6 Secure voice. The IG shall provide the means for system secure voice capability. This provision shall be accomplished by interfacing the CV-3333 Audio Converter (Vocoder) digitized voice data with the TSEC/KG-36 Crypto and the TD-1271 demand assigned multiple access (DAMA) modem.

3.5.1.7 Loopback and BITE testing. The IG shall provide CPU controlled testing to isolate system and component faults. The capability to perform system transmit and receive tests through the use of a loopback mode shall be included. In this mode a test signal shall be transmitted and received over a designated path. The loopback mode shall include crypto loop test, voice loop test, and DAMA loop test. In addition, the IG shall have BITE to assist in the detection and isolation of IG fault and interface failures.

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3.5.2 Signal flow and internal functions. The IG shall be divided into two general areas. One area shall provide processing of RED (classified nonenciphered) data and the other shall provide processing of BLACK (enciphered) data. Units that will interface with the RED data area of the IG are the vocoder (CV-3333), DAMA MODEM (TD-1271( )) and the NAVMAC/CUDIX (AN/UYK-20). The RED I/O port of the TSEC/KG-36 crypto shall also interface with the IG RED data area. Units that shall interface with the BLACK data area of the IG are the TD-1271( ) DAMA MODEM and the BLACK I/O port of the crypto. All signals in or out of the IG shall be controlled at the control panel.

3.5.2.1 Basic transmit flow. For system transmission the IG shall receive signals input from the TD-1271( ), AN/UYK-20 or CV-3333. A transmit (XMT) input signal shall initially enter a DART card (TD-1271( ) RRI, vocoder or CPTTY DART) or NTDS card for processing and interfacing with a common data bus. The bus shall connect these signals with an internal data processor consisting of a CPU, ROM, and RAM. Through programming, timing, and control, the processor shall transfer the signals to the crypto RED data input via the crypto DART. After the data are enciphered by the crypto, the data shall be fed back to the IG appearing as BLACK data at the BLACK DART. The RIB-control circuit card shall then feed the BLACK data to the TD-1271( ) for transmission.

3.5.2.2 Basic receive flow. When the transceiver receives signals (RCV) designated for the vocoder, AN/UYK-20 or TD-1271( ), the transceiver forwards these signals (RCV) to the RIB card. The signals shall then follow a reverse flow from that of the XMT signals. The RIB shall forward the signals to the BLACK data input of the crypto through the BLACK DART. The signals shall be deciphered by the crypto and fed out as RED data to the crypto DART, then through the data bus to the specific DART or NTDS card for the interface with either the vocoder AN/UYK-20 or TD-1271( ).

3.6 Human engineering. The contractor shall comply with the human factors operational and maintenance design criteria of MIL-STD-1472 to achieve effective man machine integration for both operation and maintenance.

3.7 Safety. Safety shall be in accordance with the Safety (personnel hazard) paragraph of MIL-E-16400.

3.7.1 Leakage current. IG leakage current shall not exceed 5 milliamperes.

3.8 Physical characteristics. Physical characteristics shall be as specified in 3.8.1 and 3.8.2.

3.8.1 Dimensions. The IG shall permit mounting in a standard 19-in. rack as specified in EIA RS-310-C-77. Dimensions shall conform to DWG 55910-0104150.

3.8.2 Weight. The IG weight, including the control panel, shall not exceed 73 pounds (lbs). The IG shall be prominently labeled in accordance with the weight labeling requirements of MIL-STD-1472.

3.9 Environmental service conditions. The IG shall be capable of continuous reliable operation under the environmental conditions specified in 3.9.1 through 3.9.9. The IG shall operate within tolerances throughout the tests specified herein, without alignment or adjustment, other than accessible controls on equipment front panels.

3.9.1 Temperature. Temperature shall be as specified in 3.9.1.1 and 3.9.1.2.

3.9.1.1 Operating temperature. The equipment shall conform to the requirements specified herein when operating over a temperature range of -0° C to +50° C.

3.9.1.2 Nonoperating temperature. The equipment shall not be damaged nor shall specified performance be degraded when restored to the operating temperature range after being subjected to the nonoperating temperature range of -62° C to +71° C.

3.9.2 Altitude (nonoperating). The IG shall conform to the requirements of MIL-STD-810, Method 500.2, Procedure I.

3.9.3 Shock. The IG shall conform to the requirements of the Shock paragraph of MIL-E-16400.

3.9 Vibration. The IG shall conform to the Vibration paragraph of MIL-E-16400. The upper vibration frequency limit shall be 50 Hz.

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3.9.5 Magnetic field. The IG shall conform to the magnetic field environmental requirements of DoD-STD-1399, Section 070, Part 1, Interface Characteristics paragraph.

3.9.6 Inclination. The IG shall conform to the requirements Inclination tests limits (submarine installation) paragraph of MIL-E-16400 to a maximum angle of 45 degrees.

3.9.7 Dust (fine sand). The IG shall conform to the requirements of MIL-STD-810, Method 510.2 and the Dust (fine sand) paragraph of MIL-E-16400.

3.9.8 Humidity. The equipment shall maintain the specified performance when exposed to relative humidity of 95 percent for continuous and intermittent periods, including conditions wherein condensation takes place in and on the equipment in the form of both water and frost.

3.9.9 Salt fog. The IG requirements for salt fog shall be in conformance with the Sheltered equipment paragraph of MIL-E-16400.

3.10 Electromagnetic environmental effects (E^3). The IG shall conform to the E^3 requirements specified in 3.10.1 through 3.10.4.

3.10.1 Electromagnetic interference (EMI). The IG shall conform to the requirements of MIL-STD-461, for Class A4 equipments when tested in accordance with the test methods of MIL-STD-462 for digital equipment.

3.10.2 Electromagnetic compatibility (EMC) guidance. MIL-HDBK-237 may be used for EMC management guidance and MIL-HDBK-241 may be used for EMI reduction in power supplies.

3.10.3 Compromising emanations (TEMPEST). The IG shall be capable of withstanding test of NACSIM 5100A and NACSIM 5110 to include electrical field test, magnetic field test, power line conduction test and signal and control line conduction test.

3.10.4 Electromagnetic pulse (EMP). All equipment interface pins and terminals of leads, including grounds and neutrals, shall not exhibit any permanent malfunction, degradation of performance, or deviation from specified indications after being subjected to a test signal having either the waveform and common mode current level shown on FIGURE 3 or the waveform and common mode voltage level shown on FIGURE 4, whichever occurs first, at the specific test frequency.

3.11 ID and marking. The IG shall be identified and marked in accordance with MIL-STD-454, Requirement 67.

3.12 Government-furnished property (GFP). The GFP specified in 3.12.1 and 3.12.2 is available to the contractor.

3.12.1 Software. The Government shall provide software and a ROM board (see DWG 55910-0104160) for programming the DART or ROM boards which includes a program for the BITE.

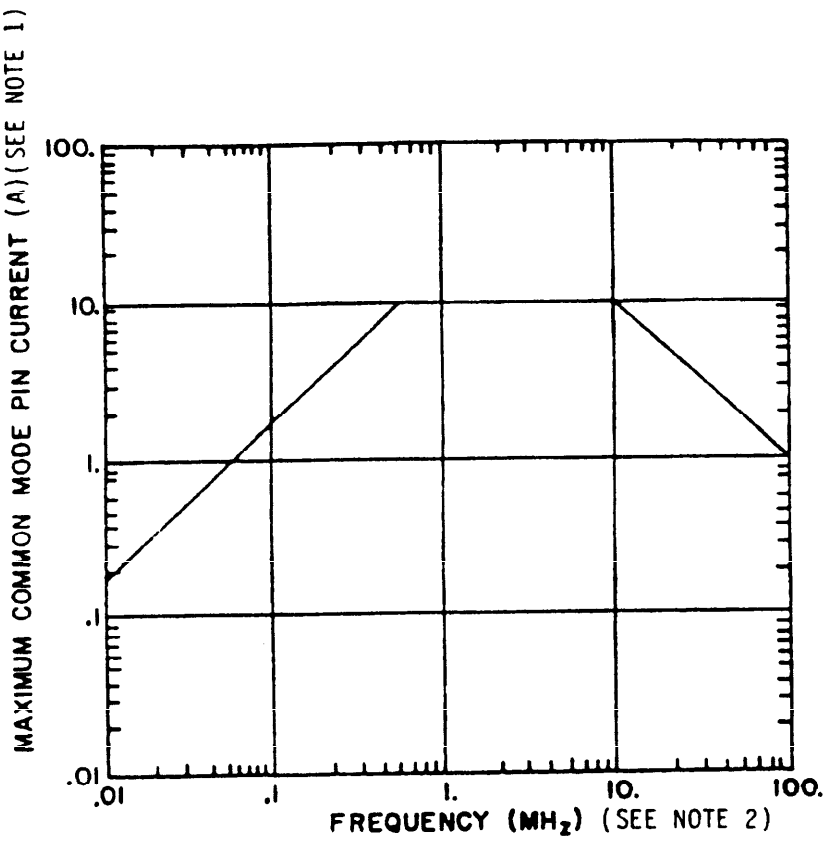
3.12.2 Equipment. The Government shall provide the equipment specified in a through e:

a. 2 crypto simulators
b. 1 TTY (AN/UGC-20 or equivalent)
c. 1 set of master patterns for PC boards
d. DWG 55910-0104160
e. 1 IG

3.13 Quantitative reliability requirement. The upper test mean-time-between-failures (MTBF) (O_0 as defined by MIL-STD-781) shall be 4000 hours for the IG. The lower test MTBF (O_1 as defined by MIL-STD-781) shall be 2000 hours. The MTBF times shall apply for Category 3A, sheltered shipboard equipment, under the conditions specified herein.

3.14 Maintainability. Maintainability shall be as specified in 3.14.1 through 3.14.3.

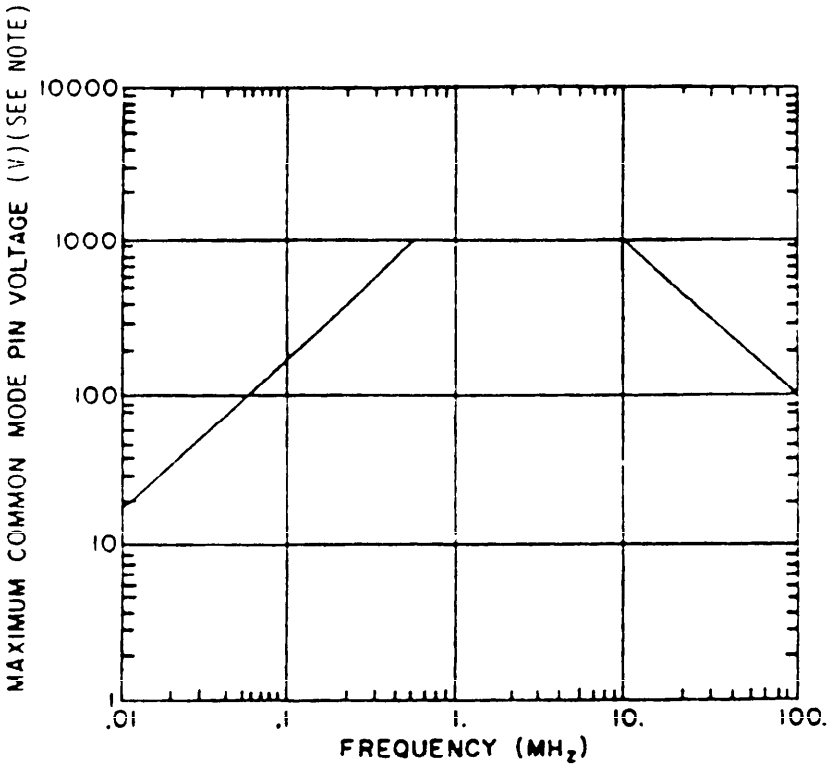
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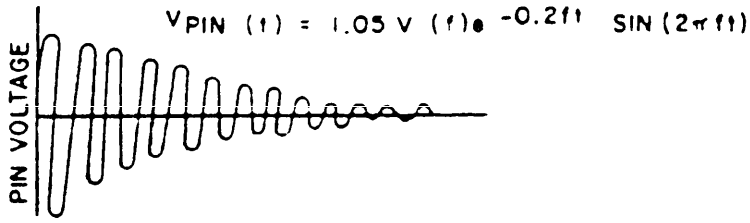


NOTES:
1. Amperes
2. Megahertz

FIGURE 3. Interface PIN current.

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NOTE: Volts

FIGURE 4. Interface PIN voltage.

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3.14.1 Quantitative organizational corrective maintenance requirement. The mean-time-to-repair (MTTR) shall not exceed 30 minutes and the maximum-corrective-maintenance-time (M_{maxct}) at the 90th percentile shall not exceed 90 minutes when corrective maintenance is accomplished at the organizational level by the replacement of lowest subassemblies (modules, printed circuit boards) and chassis-mounted parts and components (this includes electronic, electrical, electromechanical, and mechanical parts). The corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all maintenance tasks. The IG is considered in a corrective maintenance mode whenever a failed condition or stoppage of any individual operation occurs.

3.14.2 Preventive maintenance during missions. The only preventive maintenance shall consist of cleaning the IG air filter, cleaning control panel touch switches, and inspecting the IG interior. The design shall facilitate the removal and installation of the air filter so that the filter can be removed, cleaned, and installed within an elapsed time of no more than 12 minutes. Access to the RED and BLACK card cages shall be such that the cabinet case can be opened, circuit cards and wiring harnesses inspected, and the cabinet closed within 6 minutes.

3.14.3 Preventive maintenance between missions. The IG shall be designed to have a maximum-preventive-maintenance downtime of 4 hours between missions, with no single task requiring more than 2 hours.

3.15 Workmanship. Workmanship shall be as specified in 3.15.1 and 3.15.2.

3.15.1 General workmanship. Workmanship on all items manufactured under this specification shall be in accordance with the Workmanship paragraph of MIL-E-16400.

3.15.2 Workmanship screen. All production equipment shall withstand a defect detection vibration screen of random type vibration at $0.04g^2/Hz$ ±3 dB from 80 Hz to 350 Hz, and temperature cycling with operation at the maximum and minimum operating temperatures and dwell time at the nonoperating temperature extremes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
   1. Production inspection, (Group A) (see 4.4.1)
   2. Production control inspection, (Group B) (see 4.4.2)
   3. Environmental inspection, (Group C) (see 4.4.3)
c. Inspection of preparation for delivery (see 4.9).

4.3 First article inspection. Unless otherwise specified (see 6.2), the number of units specified in the contract shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE III.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4.

4.4.1 Production inspection (Group A). Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examinations and testing as will prove the workmanship and reveal the omissions and errors of the production process, such as: functional and performance tests at a limited number of points; tests which detect deviation from design; tests of adjustment; and tests which detect hidden defects of material. Production inspection shall include the tests and examinations in TABLE III, Group A.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis in conformance with the inspection procedures of MIL-STD-105 to an acceptable quality level of 6.5 percent, using the special inspection level S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. Production control inspection shall consist of: the examination, and functional and performance tests throughout the entire range of operation; detection of any deterioration of the design by wear of dies, molds, and jigs, by variations in parts; the recording of deviations in the processing temperature rise during equipment operation and ability of the equipment to withstand this heat; tests of efficiency; and tests of the system performance. These tests shall be performed on the complete equipment as offered for delivery. Production control inspection shall include the examinations and tests specified in TABLE III, Group B.

4.4.3 Environmental inspection (Group C). Environmental inspection shall encompass: tests of equipment durability; simulated service tests; tests of the effects of environment (such as extremes of temperature and humidity, effect of salt air); and effects of shock, vibration, and inclination. Environmental inspection shall include the examinations and tests specified in TABLE III, Group C. Environmental inspection shall be performed on a sampling basis on units which have been subjected to and passed production inspection.

4.4.3.1 Environmental inspection samples. Samples for environmental inspection shall be selected without regard to their quality in accordance with a and b:

a. One sample from the first 50 units fabricated
b. One sample from each successive 100 units, or fraction thereof, produced

4.4.4 Noncompliance. If a sample fails to pass Group C inspection, the manufacturer shall notify the contracting officer and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the contracting officer, has been taken. After the corrective action has been taken, Group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the contracting officer). Group A and Group B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the Group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the contracting officer.

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TABLE III. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Interface test	3.3.2	4.6.3, 4.6.4	X	X		
Steady-state voltage and frequency	3.3.2.2	4.6.1	X		X	
Transient voltage	3.3.2.2	4.6.1	X			
Transient frequency	3.3.2.2	4.6.1	X			
Spike voltage	3.3.2.2	4.6.1	X			
Power interruption	3.3.2.2	4.6.6	X		X	
Power and power factor	3.3.2.2	4.6.6	X		X	
Harmonic current	3.3.2.2	4.6.1	X			
Enclosure	3.3.4.1	4.6.1	X		X	
Thermal design	3.3.6	4.6.5	X			
Maintainability demonstration	3.14	4.7	X			
Preperformance	3.5	4.6.1	X		X	X
Safety	3.7	4.6.1	X	X		
Temperature (low and high)	3.9.1	4.6.1	X			X
Altitude	3.9.2	4.6.1	X			
Shock	3.9.3	4.6.8	X			X
Vibration	3.9.4	4.6.9	X			X
Magnetic field	3.9.5	4.6.1	X			
Inclination	3.9.6	4.6.1	X			X
Dust (fine sand)	3.9.7	4.6.1	X			
Airborne and structure-borne noise	3.3.5	4.6.1	X			X
Leakage current	3.7.1	4.6.7	X			
Humidity	3.9.8	3.6.1	X			
Salt fog	3.9.9	4.6.1	X			
Fungus	3.2.3.10	4.6.1	X			
EMI	3.10.1	4.6.10	X			X
TEMPEST	3.10.3	4.6.11	X			
Surface examination						
Parts and material	3.2	4.6.1	X	X		
Size	3.8.1	4.6.1	X	X		
Weight	3.8.2	4.6.1	X	X		
Weight labeling	3.8.2	4.6.1	X	X		
Finish	3.2.4.4	4.6.1	X	X		
Marking	3.11	4.6.1	X	X		
Workmanship	3.15	4.8	X	X		
EMP	3.10.4	4.6.10.1	X			
Workmanship screen	3.15.2	4.8.2		X		
Interchangeability demonstration	3.2.1.2	4.6.2.1, 4.6.2.2	X			

4.5 Test performance. Test equipment and test conditions for first article inspection and quality conformance inspection shall conform to the quality assurance provisions of MIL-E-16400 specified in a through c:

a. Test facilities, chambers, and apparatus
b. Standard test conditions
c. Performance of test

4.6 Test methods. Test methods shall be as specified in 4.6.1 through 4.6.11.

4.6.1 Tests and examinations. Tests and examinations specified in TABLE III shall be conducted in accordance with the applicable test methods of MIL-E-16400, except as specified in 4.6.2 to 4.6.11.

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4.6.2 Interchangeability tests. Interchangeability tests shall be as specified in 4.6.2.1 and 4.6.2.2.

4.6.2.1 RAM and DART board interchangeability test. The IG shall be tested with each of the four DART boards in each slot provided for DART boards. That is, install DART board 1 in slot 1, DART board 2 in slot 2, DART board 3 in slot 3, and DART board 4 in slot 4. Test the equipment for performance. Repeat the test with DART board 1 in slot 2, DART board 2 in slot 3, DART board 3 in slot 4, and DART board 4 in slot 1. Continue testing until each board has been tested in each slot. Perform the same test for the three RAM boards in the three slots provided.

4.6.2.2 LRU interchangeability. Each IG LRU manufactured under this specification shall be tested for interchangeability with the Government-furnished LRUs. The interchangeability test shall be conducted by interchanging the new LRUs with those of the Government-furnished IG in every combination, and testing each IG for proper operation. The test shall be conducted in accordance with a contractor-furnished and Government-approved interoperability test plan.

4.6.3 Interface test. The BITE testing shall be conducted with the interfacing equipment shown in FIGURE 1 to demonstrate compliance with 3.3.2.

4.6.4 CUDIX/NAVMACS data link. The CUDIX/NAVMACS data link shall be tested by BITE.

4.6.5 Thermal design. The thermal performance test of MIL-E-16400 shall be performed while the equipment is undergoing the first article temperature test.

4.6.5.1 Thermal survey. A thermal survey shall be performed on parts dissipating 1 percent, or more, of the total power dissipation of the unit in which the part is located and to ensure that heat sensitive parts are within the thermal stress specified in the reliability prediction report. The thermal data shall include but not be limited to the actual operating temperature of the part and any visible evidence of deterioration of the equipment.

4.6.5.2 Survey conditions. The survey shall be performed in accordance with the Test conditions paragraph of MIL-E-16400.

4.6.6 Electrical power tests. Tests shall be run in accordance with DoD-STD-1399, Section 300, to ensure compliance with the requirements of 3.3.7. Loss of memory contents is acceptable during power interruption testing.

4.6.7 Leakage current test. Leakage current shall be tested as specified in 4.6.7.1 and 4.6.7.2.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES WITHOUT ADEQUATE ELECTRIC SHOCK PROTECTION.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.6.7.1 IG test connections. After power removal each IG shall be placed on an insulated surface. The safety ground conductor between the IG and the source power shall be opened during the test. The IG shall be connected as shown in FIGURE 5, if it is connected to single-phase source power.

4.6.7.2 Measurement. Leakage current shall be measured on IG in its normal operating configuration. IG controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the higher and lowest nominal power line frequencies for which the IG is designed. The overall measurement error shall not exceed five percent. The probes shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed. The voltage shall be measured from each part of ground and from each part to all other simultaneously accessible parts for every combination of switch positions available in FIGURE 5. The open safety ground conductor shall be reconnected immediately after the test is completed.

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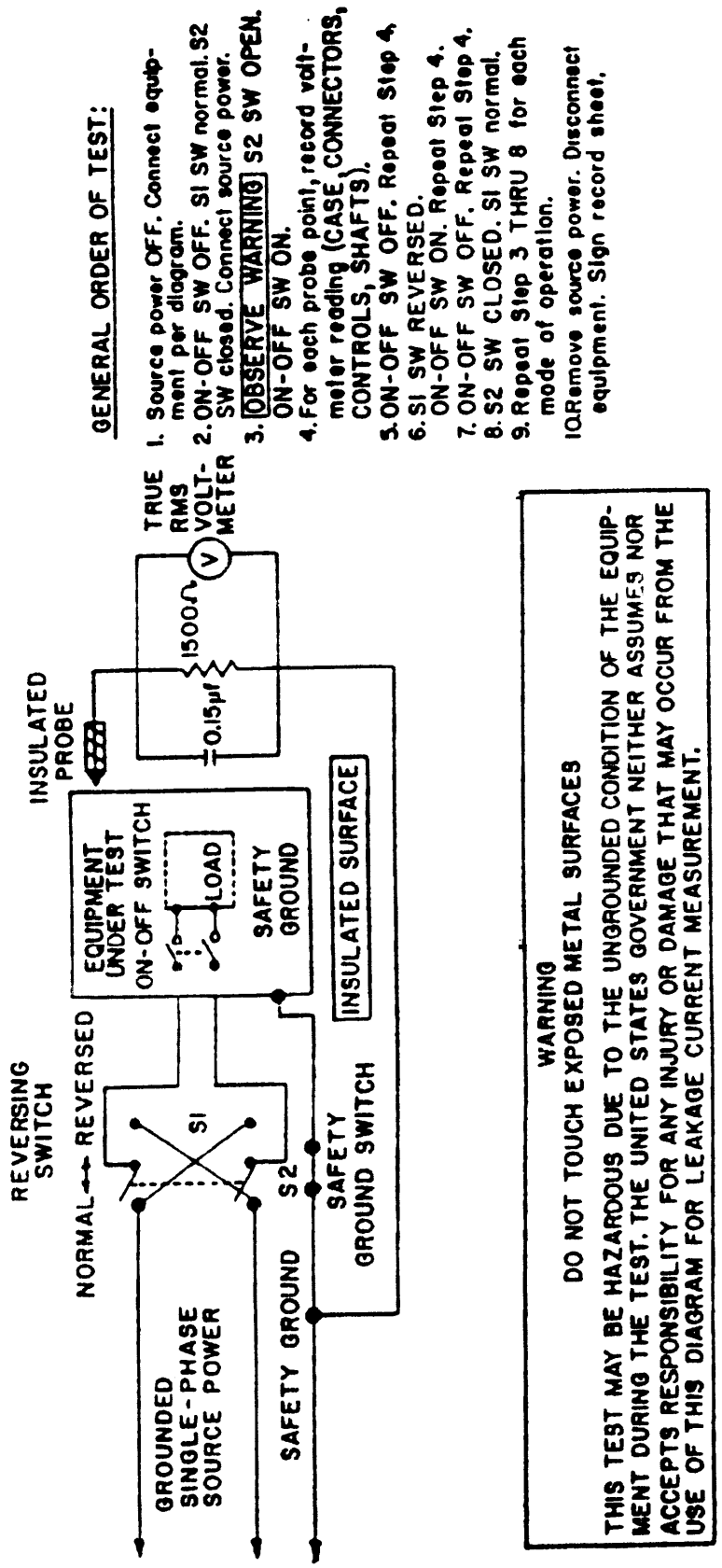


FIGURE 5. Single-phase test diagram for leakage current measurement.

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4.6.8 Shock. The IG shall be shock tested in accordance with the Shock paragraph of MIL-E-16400 to determine conformance to the requirements of 3.9.3. The unit under test shall be operated before, during, and after the shock test. It is acceptable that circuit breakers may be tripped or that some loss of memory contents results from the shock impact, however, the IG shall be capable of normal operation after re-initialization procedures following the shock test.

4.6.9 Environmental vibration. The IG shall be subjected to the test specified in the Environmental vibration paragraph of MIL-E-16400 to determine compliance to 3.9.4.

4.6.10 EMI tests. Conformance to EMI emission, and susceptibility requirements shall be verified by tests performed in accordance with MIL-STD-462.

4.6.10.1 EMP test. Conformance to the requirements of 3.10.4 shall be verified by the test method specified in a through d:

a. Applicable test frequencies. The tests shall be performed for the injection frequencies of 0.01 MHz, 0.1 MHz, 0.5 MHz, 1 MHz, 2 MHz, 5 MHz, 10 MHz, 20 MHz, 50 MHz and 100 MHz. In addition, the test shall be performed at the test sample's critical frequencies, such as local oscillator, power switching frequencies and harmonics, radio frequencies, clock frequencies, bandpass frequencies, and so forth, as specified in the EMI test plan.

b. Apparatus. Test apparatus shall consist of the items specified in 1 through 4:

1. Source injection pulse generator with the capabilities specified in A through D:

A. Damped sinusoidal transient as shown in the applicable limit and as described by the equations specified in (1) and (2):

(1) $I_{pin(t)} = 1.05\ I(f)e^{-0.2\ ft}\ SIN\ (2\pi\ ft)$

(2) $V_{pin(t)} = 1.05\ V(f)e^{-0.2\ ft}\ SIN\ (2\pi\ ft)$

where:

$I_{pin(t)}$ = Pin to case current, in A at time, t.

$V_{pin(t)}$ = Pin to case voltage, in V at time, t

f = Specific test frequency, in MHz

t = Time, in seconds

I(f) = Current determined from FIGURE 3

V(f) = Voltage determined from FIGURE 4

B. Variable amplitude so the injected transient can be increased from a minimum to a maximum specified level (that is, at least 10 percent, 50 percent, and 100 percent of the amplitude must be tested at each specified test frequency)

C. Frequency range capable of generating the required test frequencies

D. Current output level of at least 10 A into a 100-ohm load

2. Current probe

3. Oscilloscope with a minimum bandwidth capability of 100 MHz

4. Interconnecting test cable

c. Test setup and procedure. The test setup and procedure shall be as specified in 1 through 3:

1. Functional analysis. A functional analysis shall be performed before the certification testing to identify the items specified in A through C:

A. The interface pin connectors

B. The test sample functions which must be monitored to determine system failure or transient upset

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C. The additional test frequencies (that is, interface cable resonant frequency, clock frequencies, data rates, and so forth)

2. Test setup. The test setup shall be in accordance with FIGURE 6.

3. Test procedure. The test procedures shall be as specified in A through E:

A. Apply the test signal to each interface pin of the test sample, at each test frequency

B. Monitor the test sample functions in all modes of operation to determine whether there is a failure or transient upset

C. Inject one transient pulse on each interface pin and at each test frequency

D. The test sample shall be tested with its power off and on

E. Record test frequency and level at which a failure or upset occurs.

d. Notes:

1. The test sample shall be tested on a ground plane. There shall be no insulated materials between the test setup and the ground plane.

2. The test cable running between the test sample and the damped sinusoidal generator shall be kept 4 centimeters (cm) to 6 cm above the ground plane.

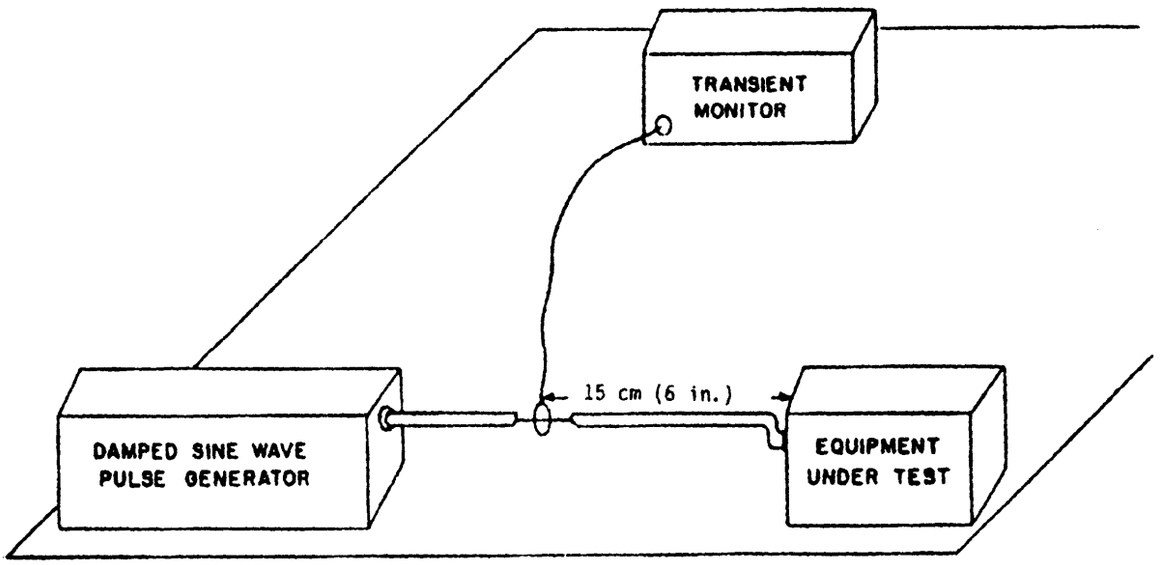


FIGURE 6. EMP test setup.

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4.6.11 TEMPEST. Profile TEMPEST tests shall be performed to demonstrate compliance with NACSIM 5100A and NACSIM 5110 requirements. These tests shall consist of at least one each electric field test, magnetic field test, power line conduction test, and signal and control line conduction test.

4.7 Maintainability demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be performed by qualified technicians, and shall be used to verify that the equipment corrective maintenance requirements are fulfilled.

4.7.1 Demonstration conditions. The equipment maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual printed circuit boards) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. The contractor shall identify 20 candidate corrective maintainability tasks (determined in accordance with APPENDIX A of MIL-STD-471). The information specified in a through c shall be made available for each candidate task:

a. Designation of specific faulty part
b. Failure mode
c. Means of introducing fault (substitution of faulty part or simulation thereof).

The maintainability demonstration shall be conducted under the standard test conditions of MIL-E-16400. The contractor shall use only test equipment, tools, jigs, fixtures, software programs, and technical documentation used in the manufacture and checkout of this equipment as specified in the maintainability demonstration plan.

4.7.2 Accept or reject criteria. The accept or reject criteria for the demonstration of the corrective maintenance times shall be as specified in TABLE IV:

TABLE IV. Acceptance criteria.

	M_{ct} 1/	M_{maxct}
Acceptance level	8	0
Sample size	20	20

1/ Mean-corrective-maintenance-time.

Acceptance shall occur when the number of observed corrective maintenance task times, which exceed the required value for each index (M_{ct}, M_{maxct}), is less than or equal to that specified in TABLE IV corresponding to each index, for the specified confidence level. The duration of each task shall be compared to the required value(s), and determined as greater or less than each index value. The actual value obtained shall also be determined. An acceptance decision can be made only when the values obtained for both M_{ct} and M_{maxct} have been accepted. The accept or reject criteria for preventive maintenance are based on the capability to perform each preventive maintenance task with no degradation in IG performance. Occurrence of IG performance degradation shall be cause for rejection of a prevertive maintenance task. The demonstration shall be performed by a technician with experience equivalent to a Navy ET-2.

4.8 Workmanship. Workmanship shall be as specified in 4.8.1 through 4.8.2.3.

4.8.1 General workmanship. The equipment, including subassemblies and assemblies, shall be examined for workmanship and soldering during fabrication and assembly for conformance to the requirements of 3.2.4.1 and 3.15.

4.8.2 Workmanship screen. Vibration and temperature cycling shall be performed on each equipment as specified in 4.8.2.1 and 4.8.2.2. Vibration shall be performed prior to temperature cycling.

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4.8.2.1 Random vibration. The vibration may be performed at the module, drawer, or end item level. All the hardware including cables and connectors shall be exposed to vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-randum or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 7. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz, and 100 Hz, for vibration frequencies above 500 Hz, shall be used for the control and analysis of the acceleration spectral density (ASD). The instantaneous acceleration peaks may be limited to three times the rms acceleration level. The item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

4.8.2.2 Temperature cycling. Each unit shall be subjected to 10 cycles of the temperature curve shown in FIGURE 8. The temperature rate of change shall be not less than 5° C per minute. Unit power shall be turned on and off at the indicated times. The unit shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the unit reaches 80 percent of the chamber temperature. When failures occur, the unit shall be reworked and the cycling continued for a cumulative total of 10 cycles.

4.8.2.3 Chassis test. After the chassis has been completely fabricated (not including the PC cards) conduct a continuity or short test on all possible combinations of connection points simultaneously, that is, card cages, rear panel connectors, RED/BLACK interface, power supply connections, and so forth.

4.9 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The IG covered by this specification is intended to provide for satellite communications within one of three networks: satellite link communications via the TD-1271( ) DAMA modem, or the FLTSATCOM secure voice net, or a nonDAMA NAVMACS/CUDIXS satellite link.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Strapping options required (see 3.3.2.4)
c. Quantity of first article ON-143(V)6/USQs to be furnished (see 4.3)
d. Levels of preservation, packaging, packing, and marking (see 5.1).

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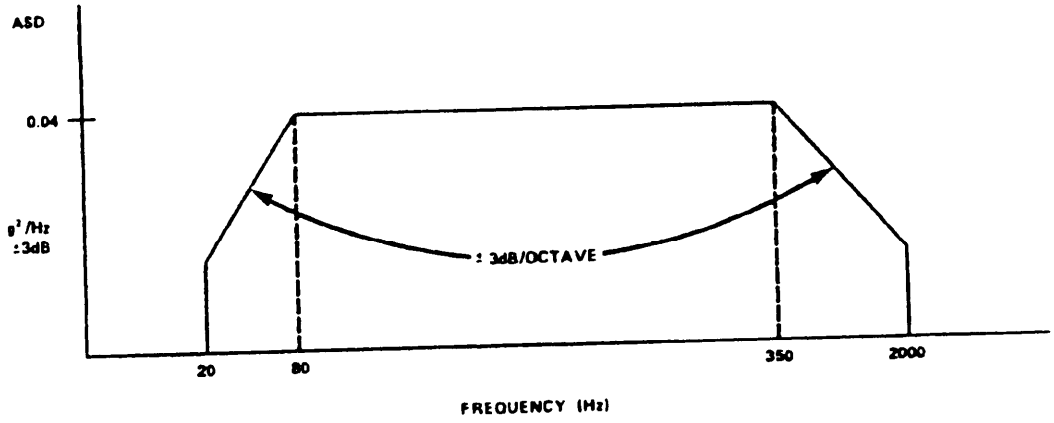


FIGURE 7. Random vibration curve.

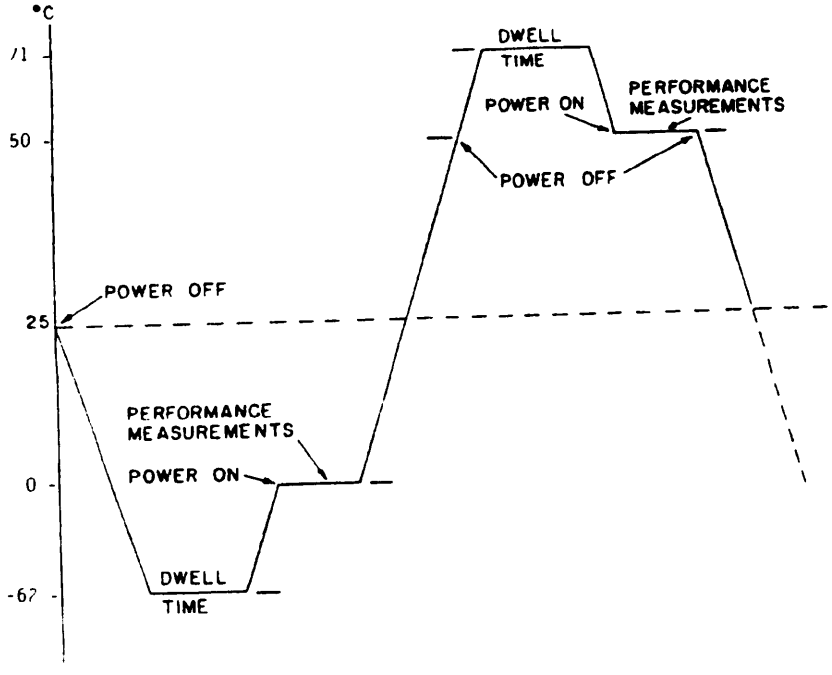


FIGURE 8. One cycle of temperature curve (not to scale).

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6.3 First article. When a first article inspection is required, the item(s) should be a first production item. The first article should consist of the number of units specified in the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.12.

6.5 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog. (Copies of this catalog may be consulted in the office of the Defense Contract Administration Service.)

6.6 Subject terms (key word) list.

Altitude
AN/UGC-20(TTY)
AN/UYK-20
Audio, Converter (Vocoder), CV-3333
Built-in test (BIT)
Built-in-test equipment (BITE)
Corona and electrical breakdown prevention
Crypto, TSEC/KG-36
CV-3333, Audio Converter (Vocoder)
Demand assigned multiple access (DAMA)
Derating
Distribution statement A
Dust (fine sand)
Dynamically adaptive receiver transmitters (DARTS)
Electromagnetic compatibility (EMC)
Electromagnetic environmental effects (E^3)
Electromagnetic inteference (EMI)
Electromagnetic pulse (EMP) enclosures
Fleet Satellite Communications (FLTSATCOM) Secure Voice
Fungus
Harmonic current
Human engineering
Humidity
Interconnecting
Inclination
Interchangeability Group, ON/143(V)6/USQ
Interconnecting Group, ON-143(V)9/USQ
Interface RED/BLACK
Leakage current
Link functions
Lubricants
Magnetic field
Maintainability
Mean-time-between-failures (MTBF)
Modular construction
NAVMACS
NTDS (Naval Tactical Data Systems)
Noise, airborne and structureborne
Nonstandard parts
Officer-in-Tactical Command Information Exchange Subsystem (OTCIXS)
ON-143(V)6/USQ, Interconnecting Group
Power and power factor
Power interruption
Prime power design
Random vibration
RED/BLACK interface
Reliability

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Safety
Salt fog
Shock
Software
Soldering
Spike voltage
Standard parts
Steady state voltage and frequency
TD 1270 ( )
TD 1271 ( )
Temperature
TEMPEST
Thermal design
Thermal stress
Transient frequency
Transient voltage
TSEC/KG-36 Crypto
Vibration
Welding structural
Workmanship screen

Preparing activity:
NAVY - EC

(Project No. 5820-N820(EC))

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-29503(EC)	INTERCONNECTING GROUP, ON-143(V)9/USQ

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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